

NinaivX — Technical Architecture

FastAPI + LangGraph agent · Claude (AWS Bedrock) · ElevenLabs voice · Supabase (Postgres)

CLIENT

React Native app (iOS / Android)

Records audio, plays replies. Sends a Supabase login token on every request.

↓ HTTPS + Bearer token

API LAYER — FASTAPI (APP/API/ROUTES.PY, PREFIX /API)

Auth (Supabase JWT)

app/core/auth.py — verifies token, loads user

Ownership check

is this persona yours?

Endpoints

auth (signup/login) · personas CRUD · chat · voice-chat · clone/select voice

AI disclosure (public)

GET /api/disclosure — transparency notice for onboarding + in-app indicator

↓ run_agent()

AGENT LAYER — LANGGRAPH STATEGRAPH (APP/AGENTS/GRAPH.PY)

Supervisor

routes by persona_type

Deceased node

air-gapped · NO tools

Companion node

has web_search tool

web_search tool

live results (httpx + DuckDuckGo)

Reply

back to API

↓ calls services

SERVICES LAYER (THE WORKERS)

llm_service.py

get_llm() → Claude · update_conversation_checkpoint() (memory)

voice_service.py

speech_to_text · text_to_speech · clone_voice · list_voices

↓ external calls

EXTERNAL PROVIDERS

AWS Bedrock

Claude Sonnet 4.6 — the reasoning brain

ElevenLabs

Speech-to-Text (Scribe), Text-to-Speech, Voice Cloning

Supabase (Postgres)

Users · Personas · Conversation memory · Auth

Key design decisions

- **Agent, not a fixed script:** the AI *decides* whether to use a tool (web) — the deceased persona is denied that ability in code.
- **Privacy by architecture:** air-gap is enforced by not binding tools, not by a prompt request.
- **Memory = rolling summary**, not full transcripts — small + fast.

Performance & safety

- **Low latency:** memory rewrite runs as a background task after the reply is sent.
- **Secure:** every request needs a valid Supabase JWT; personas are owner-scoped.
- **Resilient:** app boots even if a provider key is missing (clean 503).
- **Transparent:** persona never denies being an AI; honest if asked, plus a public disclosure notice.

Two request flows

- **Text chat:** /api/chat → agent → reply (JSON).
- **Voice "phone call":** audio → STT → agent → TTS → audio, in the persona's cloned voice.
- Same brain serves both.

Status: backend built & tested end-to-end ✓ · Next: front-end integration + deployment · Ashwin Kumar Ramesh Kumar — MSc AI, University of Sheffield